AimNxt
School of Tech Leaders

Gen AI + Agentic AI with Python

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About the Program

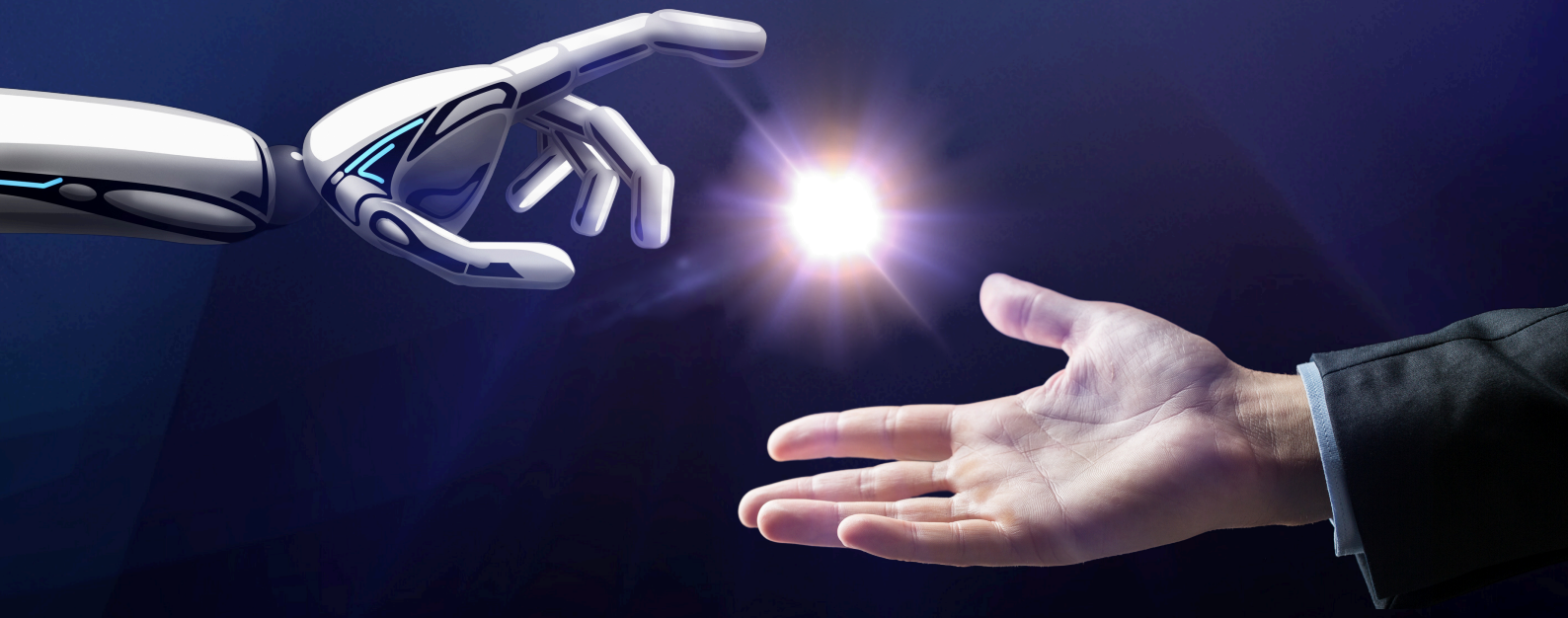
Generative AI stands at the forefront of technological innovation, revolutionizing various sectors from healthcare to finance, entertainment, and beyond. By mastering Generative AI, you will position yourself at the cutting edge of this transformative technology, opening doors to numerous career advancements and opportunities.

Why Generative AI?

Generative AI is not just a buzzword; it's a game-changer. It powers intelligent systems that can create realistic images, write compelling content, enhance conversational
agents, and even develop complex simulations. As industries across the globe integrate Generative AI into their operations, the demand for skilled professionals in this field is skyrocketing

Career Boost and Salary Hikes

Investing in Generative AI knowledge can significantly enhance your career trajectory. According to industry reports, professionals with expertise in AI and Machine Learning are seeing average salary hikes of up to 20-30%. Moreover, the growing necessity for AI-savvy leaders in senior roles, such as AI Architects or Lead Data Scientists, often comes with lucrative compensation packages that reflect their specialized skill set.



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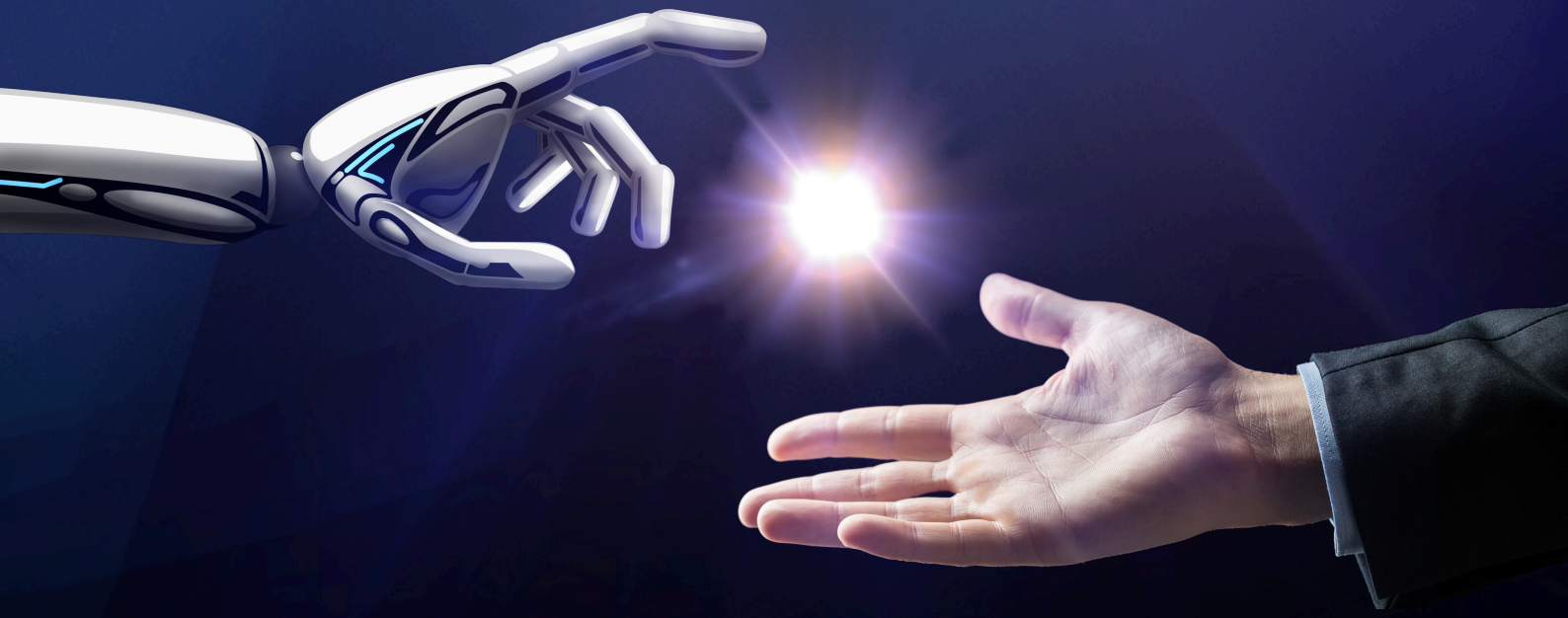


Key Features of the Course

Practical Focus:

Hands-on exercises and real-world projects to reinforce learning

Industry Relevance:

Coverage of industry best practices and real-world challenges.

Expert Instruction:

Delivery by experienced AI practitioners with deep industry knowledge

Flexible Learning:

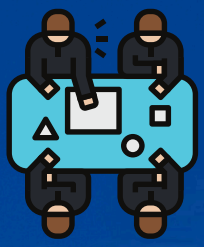
Self-paced learning with recorded sessions and interactive Q&A sessions.

Flexible Learning:

Mock Interview Questions

Career Guidance

Project at the end of every track



Capstone project at the end

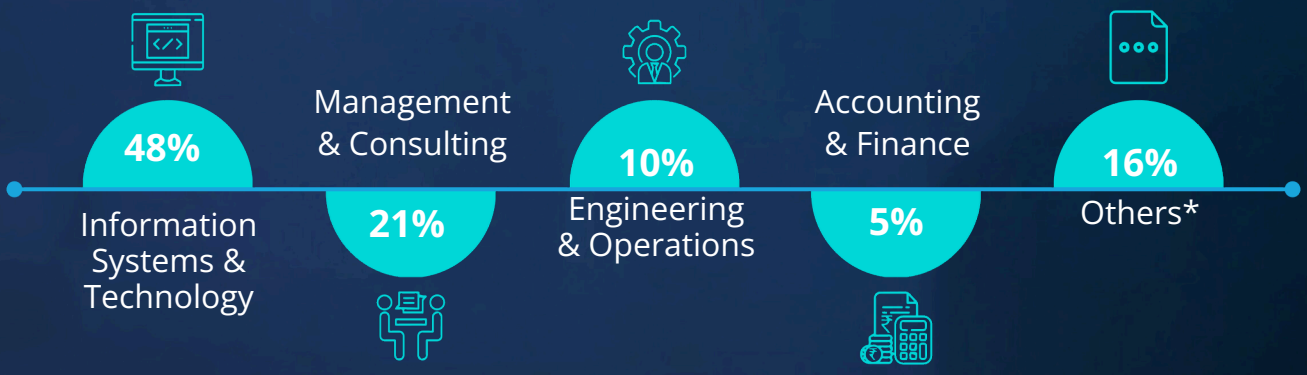
Industry Projections

The AI market is projected to reach $190 billion by 2025, with Generative AI being a substantial contributor to this growth. Companies are investing heavily in AI capabilities,
seeking professionals who can innovate and lead AI projects. By gaining proficiency in Generative AI, you position yourself at the helm of this expansive growth, ready to drive impactful changes across industries.

Transformative Potential

**Others include Manufacturing & Operations, Communications, Marketing, Sales, FMCG, Hospitality etc.*

Functions



**Others include Marketing & Sales, HR, Delivery, R&D etc.*

Curriculum Outline

TRACK 1: Setup & Toolchain

Learning Out Comes

Outcomes: Productive Python dev environment, CI-ready repo.

- Environment: conda, venv, VS Code, Jupyter.
- Packaging: pip.
- Quality: black, ruff, mypy, pytest, coverage.
- Git essentials; GitHub/Azure DevOps repo + simple CI pipeline.

Lab: Initialize repo, enforce formatting in CI, publish a small wheel.

Phase 1 — Python Foundations

Outcomes: Solid Python fluency for data + services.

Core: data types, control flow, functions, iterators/generators.

- OOP + data classes + pydantic models.
- Files, JSON/YAML, pathlib, logging.
- Typing with typing & mypy.
- Concurrency: asyncio, aiohttp, task groups.

Labs:

- CLI log parser (sync vs async).
- Typed config loader with pydantic + env overrides.

Curriculum Outline

Phase 2 — Data Structures, Algorithms & System Design (2–3 weeks)

Outcomes: Problem solving + design for scalable AI apps.

- DS/Algo refresh (lists, dicts, heaps, graphs; Big-O).
- API design: REST (FastAPI), GraphQL (Strawberry optional).
- Caching: Redis patterns, idempotency keys.
- **Storage:** PostgreSQL basics (SQLModel/SQLAlchemy), Cosmos DB intro.
- Design patterns for Python (Strategy, Factory, Adapter).
- Cloud basics on Azure: resource groups, VNet, identity.

Labs:

- FastAPI service with rate-limit + Redis cache.
- Design a URL shortener (read/write paths, cache, DB).

Phase 3 — ML/AI Foundations in Python (3 weeks)

Outcomes: ML literacy and workflow discipline.

- Data wrangling: pandas, polars (optional), EDA with notebooks.
- ML: scikit-learn pipelines, metrics, model validation.
- DL intro: torch or tensorflow (choose one), training loop anatomy.
- Experiment tracking: mlflow basics.
- Packaging + reproducibility for ML code.

Labs:

- Regression + classification pipeline with proper CV & metrics.
- Torch MLP baseline on tabular data with MLflow tracking.

Curriculum Outline

Phase 4 — Generative AI with Python + Azure AI Foundry (4–5 weeks)

Outcomes: Build real GenAI apps, safely and reliably.

- GenAI concepts: transformers, tokenization, prompting patterns.
- **Azure AI Foundry (formerly AI Studio):**
  - o Create hub/project, connect Azure OpenAI deployments.
  - o Evaluate prompts in Prompt Flow.
  - o Data connections, environments, managed endpoints.
- **Python SDKs & APIs:**
  - o openai (Azure OpenAI), azure-identity, azure-core.
  - o Embeddings + vector stores: Azure AI Search (azure-search-documents) or FAISS.
- **RAG end-to-end:**
  - o Chunking, embeddings, metadata.
  - o Query rewriting, reranking options.
  - o Evaluation (faithfulness, groundedness).
- Content Safety & Guardrails: Azure AI Content Safety SDK; red teaming basics.

Labs:

- Prompt engineering notebook with eval set in Foundry.
- Build a RAG API: FastAPI + Azure OpenAI + Azure AI Search + Blob Storage.
- Add Content Safety checks and structured output via JSON mode

Curriculum Outline

Phase 5 — Fine-Tuning & Optimization (2 weeks)

Outcomes: Tailor models and control costs/latency.

- Fine-tuning small LLMs via Azure AI Foundry pipelines (where available) or
- LoRA/PEFT (Hugging Face) hosted on Azure.
- Prompt caching, response streaming, batching.
- Latency/throughput tuning; cost observability.

Labs:

- Fine-tune a classification or style adapter; compare vs prompt-only.
- Add streaming + server-sent events to your FastAPI endpoint.

Phase 6 — Agentic AI in Python (4 weeks)

Outcomes: Design, build, and evaluate autonomous agents.

- **Agent concepts:** planning, tools, memory, reflection, multi-step control.
    - o Frameworks (Python):
    - o LangChain Agents (Tools, Toolkits, AgentExecutors).
    - o LlamaIndex Agents (QueryEngines, Tools, workflows).
    - o AutoGen (multi-agent conversations & tool use).
    - o (Optional) Semantic Kernel Python.

• **Tooling & Actions:**

    - o Web search, DB queries, internal APIs, function-calling.
    - o Structured outputs (pydantic) and schema validation.

• Long-term memory: vector stores (AI Search/FAISS), relational memory, episodic vs semantic.

• Safety for agents: tool authorization, rate limits, sandboxing, human-in-the-loop review.

• Orchestration in Azure AI Foundry Prompt Flow for multi-step pipelines

Labs:

• Research assistant agent (web + PDF tools + citations).

• Multi-agent workflow: planner, executor, reviewer; gated by human approval.

• Add memory + capabilities registry; log & replay traces

Curriculum Outline

Phase 7 — MLOps/LLMOps & Production on Azure (3 weeks)

Outcomes: Ship/operate GenAI systems reliably

- **Packaging & deployment:**
    - o FastAPI app → Azure Container Apps or AKS.
    - o **IaC**: Bicep or Terraform (RG, Key Vault, AI Search, Storage).

Secrets & identity: Managed Identity, Key Vault, role assignments.

Data: Azure Blob Storage (docs), Event Hubs/Service Bus for async.

CI/CD: GitHub Actions or Azure Pipelines (build, test, scan, deploy)

Labs:

• One-click infra deploy for the RAG service.

• Canary rollout + auto-rollback using health probes.

Phase 8 — Capstone (2–3 weeks)

Outcomes: A production-grade, demo-ready system

Pick one and scope it to your org:

1. Enterprise RAG Assistant

o Ingest SharePoint/Blob docs; Azure AI Search; citations; Content Safety; audit logs**.**

2. Agentic Ops Copilot

o Multi-agent: planner (LLM), executor (tools: Jira/GitHub/Azure), reviewer (policy).

3. Domain QA with Fine-Tuned Policies

o Mixture of prompting + light fine-tune; eval harness; safety guardrails.

Deliverables: Architecture doc, IaC, CI/CD, runbook, cost model, red-team report**.**

TRACK 2: GENERATIVE AI ESSENTIALS

Weekly Time Grid (suggested, 24–26 weeks total)

- 0: Setup (1)
- 1–3: Python foundations (3)
- 4–6: DS/Algo + system design (3)
- 7–9: ML foundations (3)
- 10–14: GenAI + Azure AI Foundry (5)
- 15–16: Fine-tuning & perf (2)
- 17–20: Agentic AI (4)
- 21–23: MLOps/LLMOps on Azure (3)
- 24–26: Capstone (2–3)

Tech Stack (Python)

- **Core:** Python 3.11+, FastAPI, pydantic, httpx/aiohttp, uvicorn.
- **ML/GenAI:** openai (Azure), langchain, llama-index, faiss-cpu, mlflow, torch (optional).
- Azure SDKs: azure-identity, azure-search-documents, azure-storage-blob,
- azure-monitor-query, azure-ai-contentsafety (or REST), azure-core.
- **Quality/Build:** black, ruff, mypy, pytest, coverage, pre-commit.
- **Infra**: Docker

Assessments & Milestones:

Checkpoints: Python quiz + coding kata; API design review; ML evaluation report; RAG evaluation (faithfulness/groundedness); agent safety test; deployment DR drill.

Rubric: Code quality, tests, observability, safety, cost discipline, and user experience.

Certifications

AI-900

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Learning Journey

Deepen Your Intellectual Capital

- World-class curriculum & teaching by leading faculty - Peer-to-peer learning

Advanced Applications

- Dive into cutting-edge tools like ChatGPT, DALL·E, and BERT-based models.
- Learn how to integrate Generative AI into real-world applications like marketing, content creation, and automation.

Enhance Your Career Capital

- Accelerate your leadership career with cutting-edge insights
- Learn while you earn and make new, life-long connections with experienced peers and industry experts

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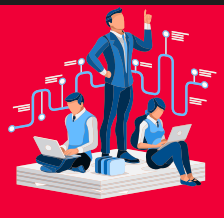
Who is this Program Ideal for?

IT Professionals

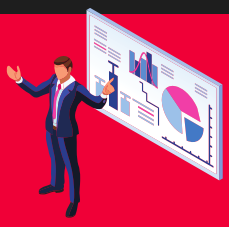
Software Developers

Data Analysts

Managers



Business Analysts



Data Engineers

Data Scientists

Beginners or recent graduates with a bachelor's or master's degree.....

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Ready to turn your dreams into reality?

Our lines are always open for dreamers!

Speak with our Career Counselor

Connect with your career counselor here:
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